01 / Preserve capital. Fund the mission.

CONDITIONAL ADJUST

USD 10 million endowment Investment committee memo

Retain the constrained minimum-variance rule as a candidate, but adjust the spending / liquidity plan and obtain approval of explicit loss budgets before implementation. Lower observed volatility does not eliminate concentrated exposures, tail risk or funding uncertainty.

Evaluation: 4 Jan 2021 - 30 Oct 2026	Observed result
Net annual growth	7.63%
Annual volatility	8.80%
Maximum drawdown	-16.78%
Current HS99 one-day ES	2.03% / $203,169
Full-history exposure HS99 ES	2.50% / $250,185

The decision in three points

Funding: the proposed 6.75% nominal hurdle is below the observed evaluation CAGR, but that historical margin is not a reliable forecast or a sequence-safe spending guarantee.

Loss budget: current HS VaR99 and ES99 marginally exceed the proposed 1.5% and 2% limits; full-history tails and crisis drawdowns are materially worse. Limits must not be relaxed merely to manufacture acceptance.

Governance: approve the mandate assumptions, the treatment of monthly weight drift and the response to clustered exceptions. Independent human review remains pending; automated checks are not a sign-off.

All figures are derived from the frozen r3 dataset and the unchanged registered portfolio rule. No strategy switch or threshold retuning follows from evaluation results. Bundle 1c7bd6007fe22b014b25.

02 / A mandate that can be evaluated

The official challenge specifies the USD 10M conservative endowment and seven-ETF universe; it leaves teams to set the horizon, objective and numerical risk budgets. The following are team-proposed planning assumptions, not facts supplied by the organiser or client.

Planning choice	Team proposal
Horizon / review	10 years; quarterly review; monthly scheduled rebalance
Spending / inflation	4.00% of beginning-year NAV / 2.50% annual assumption
Additional mandate fee	0.25%; distinct from embedded ETF fund expenses
Desired real growth	0%; preserve purchasing power after spending
Cash-flow convention	End-year spending / fee deductions measured against beginning-year NAV

Return bridge

1 + R - spending - fee = (1 + inflation) × (1 + real growth)
At zero real growth: R = 4.00% + 0.25% + 2.50% = **6.75%**. This is a planning hurdle. No probability of achieving it is asserted. Daily return paths include trading costs, but do not deduct endowment spending or this additional fee.

Proposed review budget	Fraction of NAV	USD at $10M
Daily Historical VaR95	1.00%	$100,000
Daily Historical VaR99	1.50%	$150,000
Daily Historical ES99	2.00%	$200,000
Annual volatility	10.00%	Not a one-day loss limit
Drawdown / scenario terminal loss	20.00% each	$2,000,000

Decision rule

Conditional ADJUST applies while tail budgets, income / spending feasibility, concentration or human governance remain unresolved. A review threshold is not a guarantee that losses stop there. Cash reserves, off-universe hedges and automatic daily clipping are not silently introduced.

03 / Data quality is part of the result

Input	Frozen scope / treatment
ETF market history	SPY, EFA, EEM, AGG, GLD, XLE, XLK; Yahoo Finance via yfinance
Prices / returns	5,492 NYSE prices, 3 Jan 2005 - 30 Oct 2026; 5,491 one-session returns from 4 Jan 2005
Adjusted price basis	Dividend/split-adjusted Close, auto_adjust=True; no extra dividend return added
Risk-free proxy	FRED DTB3; versioned r3 values and day-level availability; primary (quote/100)/252
Income / inflation	Separate split-adjusted dividends / non-dividend-adjusted prices; FRED CPIAUCSL context
Reproducibility	SHA-256 verified snapshots and manifests; offline load; raw inputs retained

Alignment and outliers

All seven ETFs match the expected NYSE session grid without missing price cells. The first price is an anchor, not a zero return. Missing adjacent intervals are invalidated rather than bridged or forward-filled. The original quality audit retained all 307 flags: 188 large negatives, 101 robust-MAD jumps and 18 heuristic structural-break candidates. Overlapping flags are not separate deleted observations.

Statistical conventions

Log returns support asset statistics, ADF checks and correlation diagnostics. Simple returns are used for self-financing portfolio P&L, compounding, dollar losses and tail models because these require exact arithmetic aggregation. Asset and strategy ADF tests reject a unit-root null; this does not establish independent observations or constant volatility.

Availability caveats

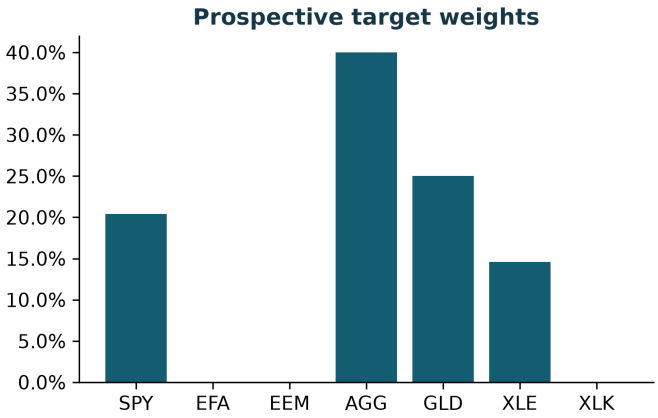
DTB3 is a discount-basis annual quote [1], not a realised trading return. r3 uses eligible observation versions as well as availability dates; older revisions cannot displace newer eligible observations. Exact intraday release hours remain unverified. There are inferred later vintages, a sparse-era calendar fallback and two unresolved observations. /252 stays primary; 91-day accrual is a sensitivity. CPI is revised ex-post context and never a trading signal.

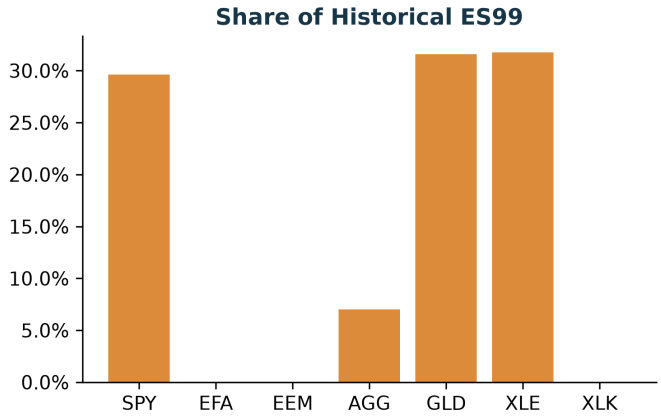
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Config SHA-256 587ddbe363589fb6956c23c5fe8b2b544bac81d9224a0b4bb59f60f5fbc0a017

04 / Construction: constraints have consequences

The unchanged rule minimises sample portfolio variance using the previous 504 sessions, sample covariance (ddof=1), long-only full investment and monthly targets. Weights used for session t are known before that session's return. Five basis points are charged per sum of absolute traded weights; net wealth compounds the cost-adjusted returns.

Target constraint	Consequence
Each ETF ≤40%	No single target dominates beyond 40%
Equity group ≤35%	At least 65% jointly in AGG and GLD
AGG / GLD implied bounds	Each 25%-40% at target; not a promise of positive returns
Defensive group ≥40%	Redundant with the stricter equity cap, retained transparently
Constraint scope	Rebalance targets only; natural drift is monitored, not clipped



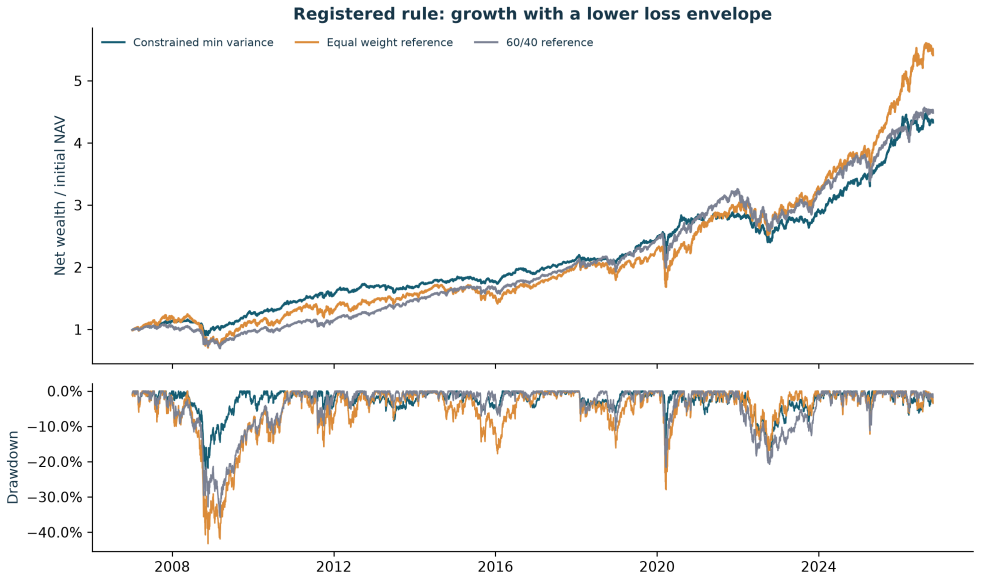


Prospective target estimated through the frozen final close. It is not the last session's pre-return holdings or an instruction already executed. The current XLE position is an optimiser outcome, not an explicit macro forecast. Rebalancing and solver records are exported.

Protocol integrity

Commit 782ec20 records the rule and proposed mandate before post-2016 strategy outcomes. Development ends in 2016; selection covers 2017-2020; evaluation begins in 2021. No selection-period winner is chosen. Prior whole-history asset EDA, RF research and fixed-exposure stress inspection were already available, so this is registered-rule evaluation rather than a pristine blinded experiment. Equal weight and 60/40 are references and may violate the candidate's mandate constraints.

05 / Results: preserve the comparison, not the story



Period / portfolio	CAGR	Volatility	Max DD	Sharpe
Development / Min variance	6.65%	8.77%	-22.18%	0.703
Development / Equal weight	5.45%	17.71%	-43.26%	0.351
Development / 60/40	6.07%	12.19%	-35.78%	0.490
Selection / Min variance	10.38%	7.74%	-14.47%	1.145
Selection / Equal weight	10.90%	15.50%	-27.94%	0.660
Selection / 60/40	11.73%	12.07%	-21.63%	0.870
Evaluation / Min variance	7.63%	8.80%	-16.78%	0.506
Evaluation / Equal weight	13.69%	12.89%	-17.49%	0.805
Evaluation / 60/40	8.41%	10.66%	-20.77%	0.502

Active paths begin 5 Jan 2007 after 504-session warmup and end 30 Oct 2026. All three use the same monthly execution / cost conventions. The constrained rule reduces volatility and drawdown relative to 60/40 over these samples, but earns less than equal weight in evaluation. We retain this trade-off rather than selecting the highest realised return after observing it. Drawdown includes initial wealth 1.0.

06 / Return is not the same as spendable income

Full active history metric	Min variance	60/40 reference
Geometric Annual Return	7.68%	7.88%
Arithmetic Annual Return	7.77%	8.27%
Annualized Volatility	8.58%	11.73%
Sharpe Ratio	0.723	0.571
Downside Deviation	6.07%	8.33%
Sortino Ratio	1.023	0.805
Capm Beta	0.290	0.590
Jensen Alpha	3.13%	0.43%
Treynor Ratio	0.214	0.114
Tracking Error	8.16%	0.00%
Information Ratio	-0.061	N/A
Max Drawdown	-22.18%	-35.78%
Calmar Ratio	0.347	0.220

CAPM / KPI interpretation

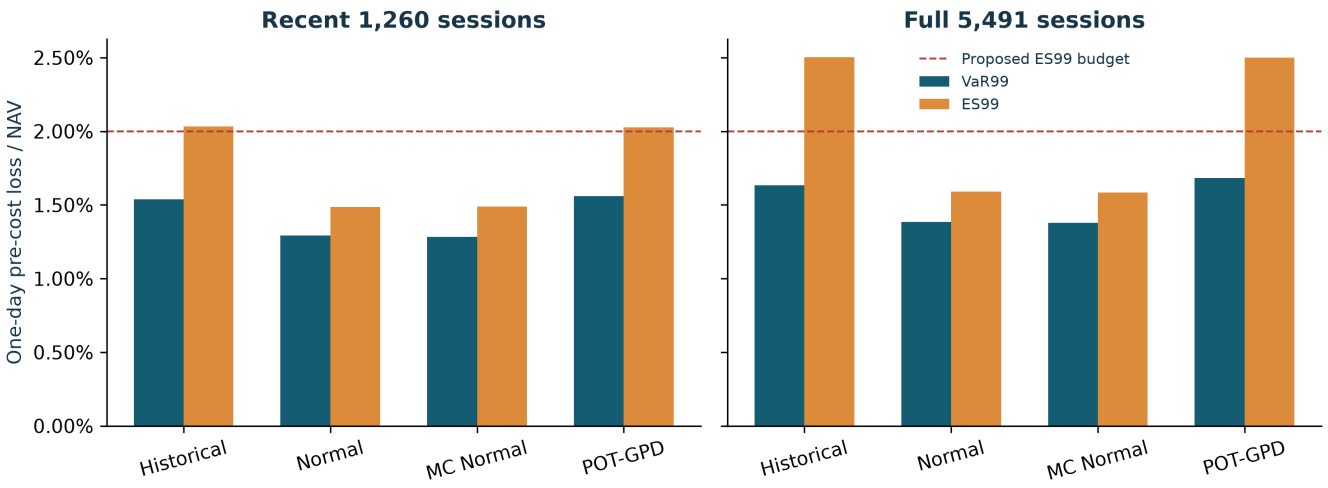
CAPM beta and Jensen alpha use SPY excess returns against day-aligned r3 RF. Tracking error and information ratio use net monthly-rebalanced 60/40. Alpha is an in-sample regression statistic, not proof of persistent skill. Ratios with zero / unidentified denominators remain unavailable. Sortino uses the RF daily target and all observations.

Income and purchasing power

The primary path’s ending trailing-12-month cash distribution yield is 2.19%, below proposed 4% spending. A pure distribution-funded policy would therefore require an adjustment; total-return spending requires asset sales and a liquidity plan. Distributions are already embedded in adjusted returns and are never added twice.

Endpoint-CPI real annual growth is 4.88% before endowment spending over 236 contiguous completed months, Feb 2007-Sep 2026. Missing October 2025 / following monthly changes do not invalidate endpoint growth, but prevent a complete intermediate monthly inflation series. October 2026 CPI was unreleased. The nominal and real intervals differ; do not subtract figures from unmatched windows. Historical purchasing-power gains do not guarantee future funding.

07 / Tail loss: Normal is a useful reference, not protection



Sample / model	Confidence	VaR	ES	ES USD
Recent / Historical	95%	0.87%	1.30%	$130,446
Recent / Normal	95%	0.90%	1.14%	$114,118
Recent / MC Normal	95%	0.90%	1.14%	$114,119
Recent / POT-GPD	95%	0.87%	1.30%	$130,454
Recent / Historical	99%	1.54%	2.03%	$203,169
Recent / Normal	99%	1.29%	1.49%	$148,667
Recent / MC Normal	99%	1.28%	1.49%	$148,814
Recent / POT-GPD	99%	1.56%	2.03%	$202,724
Full / Historical	99%	1.63%	2.50%	$250,185
Full / Normal	99%	1.38%	1.59%	$158,971
Full / MC Normal	99%	1.38%	1.59%	$158,584
Full / POT-GPD	99%	1.68%	2.50%	$250,143

All estimates use one-day pre-cost simple losses at the same prospective target weights and matched sample / confidence. Historical VaR uses an empirical quantile; ES allocates exactly the requested tail probability, including fractional / tied boundary mass. Normal VaR/ES retain the estimated mean and sample volatility. Monte Carlo uses 25,000 seeded Normal scenarios and differs through sampling, not a richer tail model.

Current recent HS99 VaR exceeds its proposed limit by $3,950 and ES99 by $3,169. These small exceedances are not statistically precise cliffs. Full-history ES99 is much higher; recent 1,260 sessions omit GFC and COVID.

08 / EVT: show where extrapolation begins

Peaks over threshold fits a Generalised Pareto distribution to strict loss excesses with location fixed at zero. The predeclared 95% threshold balances tail relevance against sample size; the threshold is not lowered to rescue a failed fit. Minimum strict excess count is 50. The empirical boundary is 1 - N_u/N, which can differ from the requested quantile because of ties.

Sample / threshold	Excesses	Shape ξ	Scale β	Status
Recent / 90.0%	126	0.113	0.00339	ok
Recent / 92.5%	94	0.093	0.00364	ok
Recent / 95.0%	63	0.054	0.00407	ok
Recent / 97.5%	31	N/A	N/A	unavailable
Full / 90.0%	549	0.193	0.00384	ok
Full / 92.5%	411	0.247	0.00370	ok
Full / 95.0%	274	0.254	0.00405	ok
Full / 97.5%	137	0.389	0.00394	ok

Diagnostics and limits

Mean residual life, shape and modified scale β - ξu are exported for the 90%, 92.5%, 95% and 97.5% candidates. Recent 97.5% has only 31 strict excesses, so its fit is unavailable. The recent 95% fit has ξ=0.054. Positive shape by itself does not reject a Gaussian model or prove a stable tail law. Fits and asymptotic standard errors do not remove volatility dependence.

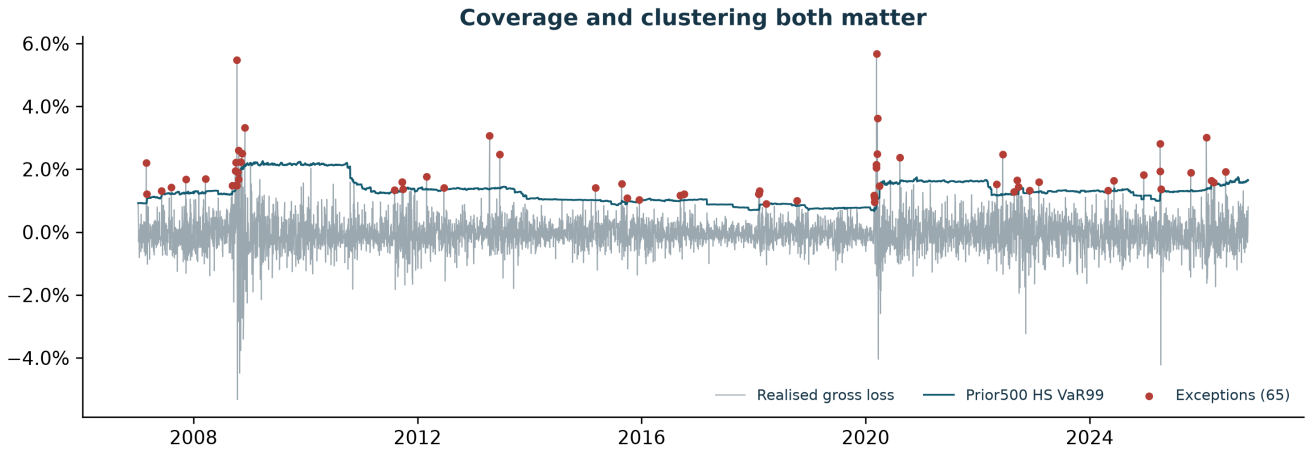
For confidence c in the fitted tail, VaR = u + β/ξ × [(p_u/(1-c))^ξ - 1], with the exponential limit at ξ = 0. ES = (VaR + β - ξu)/(1-ξ) when ξ < 1. ξ ≥ 1 makes the first moment unavailable. Negative ξ requires respecting the finite upper support. These implementation checks follow the GPD support convention [2].

Pure POT at 90% is unavailable rather than pretending that body quantiles are tail extrapolations. An explicitly separate empirical-body / GPD-tail hybrid supports body VaR and integrates its ES over the empirical body plus fitted tail. At full-history 95%, the actual boundary exceeds 95% slightly, so pure POT is unavailable there too; hybrid 95% remains labelled separately.

Development-only uncertainty evidence

Step7 includes fixed-seed moving-block bootstrap tail sensitivity and QQ/PP diagnostics. Some pure-POT 95% bootstrap intervals failed the minimum successful-replication rule and remain unavailable; hybrid intervals are separate. Those earlier development-exposure intervals are not relabelled as uncertainty bands for this new current target. No rolling EVT calibration is claimed.

09 / Backtesting: count and sequence both matter



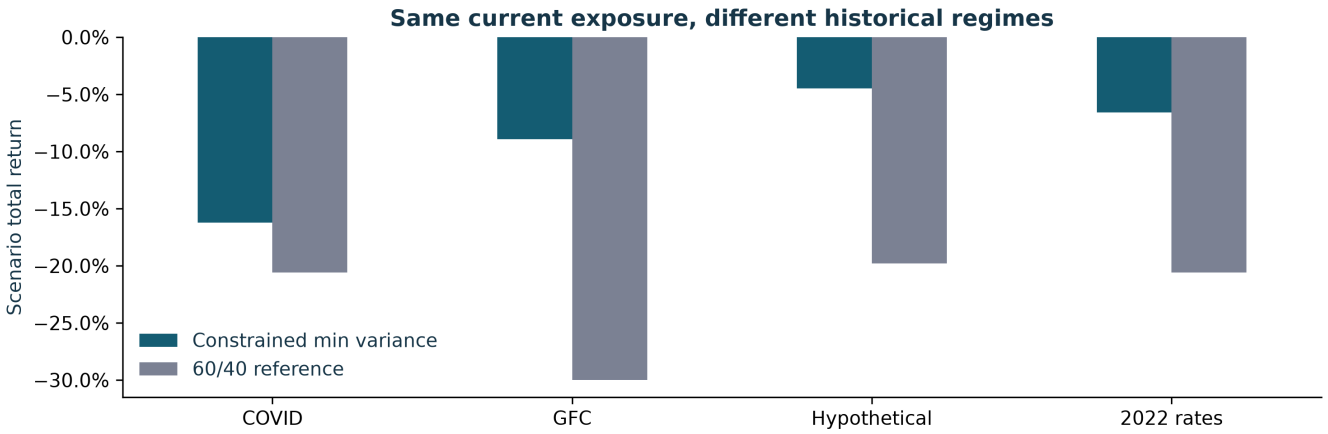
Scope / model / c	N / exceptions	UC p	IND p	CC p
Full / HS 95%	4987 / 288	0.0141	0.0001	3.4e-05
Full / HS 99%	4987 / 65	0.0397	1.7e-05	1.2e-05
Full / Normal 95%	4987 / 263	0.3792	7.1e-06	3.0e-05
Full / Normal 99%	4987 / 105	8.2e-12	2.2e-07	1.0e-16
Evaluation / HS 95%	1464 / 88	0.0850	0.0173	0.0132
Evaluation / HS 99%	1464 / 18	0.3942	0.2165	0.3233
Evaluation / Normal 95%	1464 / 74	0.9237	0.0135	0.0470
Evaluation / Normal 99%	1464 / 27	0.0037	0.0123	0.0006

Each forecast uses 500 sessions strictly before target t and the actual pre-return portfolio weights. Realised gross loss is -w□₋₁·r□. An exception requires loss > VaR (equality is not an exception). Date controls select targets without deleting warmup. HS is primary; Normal is the fixed comparison. No rolling MC or EVT results are asserted.

Kupiec tests unconditional exception frequency on all valid N. Christoffersen IND tests first-order clustering. CC combines IND with transition-conditioned UC; each contiguous run’s first state is conditioned out. Disconnected valid runs withhold CC. Asymptotic p-values, sparse cells and low power limit conclusions; non-rejection is not safety.

The final 250 sessions (3 Nov 2025-30 Oct 2026) contain **4 HS99 exceptions: green**, versus **9 Normal99 exceptions: yellow**. This educational Basel-style diagnostic [3] is not regulatory approval. It does not erase full-history HS99 frequency / independence rejection, nor Normal99 evaluation rejection. Evaluation HS99 is less concerning at these tests, while HS95 evaluation independence still rejects.

10 / Stress: a terminal recovery can hide a deep trough



Portfolio / scenario	Total return	Max drawdown	Terminal USD loss
Min variance / GFC	-8.93%	-23.87%	$892,991
Min variance / COVID	-16.23%	-18.84%	$1,622,750
Min variance / 2022 rates	-6.57%	-13.36%	$656,660
Min variance / Hypothetical	-4.48%	-4.48%	$447,652
60/40 / GFC	-29.99%	-30.33%	$2,999,420
60/40 / COVID	-20.59%	-20.81%	$2,058,822
60/40 / 2022 rates	-20.58%	-20.63%	$2,057,506
60/40 / Hypothetical	-19.80%	-19.80%	$1,980,000

Historical windows are GFC 9 Oct 2007-9 Mar 2009, COVID 19 Feb-23 Mar 2020 and rates / inflation 3 Jan-14 Oct 2022. Each replay buys and holds today’s same initial target weights through the asset history. It is an ex-post exposure diagnostic, not the historical monthly strategy. Per-asset P&L and daily paths are available in the stress engine.

The candidate’s GFC terminal loss is modest relative to its within-window drawdown. Drawdown includes starting NAV; it cannot be replaced by endpoint return. Current-target GFC drawdown breaches the proposed 20% review threshold even when terminal loss does not.

The hypothetical joint repricing shocks are SPY/EFA/EEM -25%, AGG -12%, GLD +10%, XLE +20% and XLK -30%. Its duration and probability are unassigned. Positive gold / energy shocks are assumptions, not guaranteed hedges. Never compare this one-step repricing with daily ES as if they share a horizon.

11 / From risk decomposition to a review action

ETF	Current target	ES99 share
SPY	20.39%	29.64%
EFA	0.00%	0.00%
EEM	0.00%	0.00%
AGG	40.00%	7.00%
GLD	25.00%	31.60%
XLE	14.61%	31.76%
XLK	0.00%	0.00%

Euler decomposition

Normal marginal VaR / ES include the mean term; Historical ES uses the same exact fractional tail mass as portfolio ES. Components sum to total risk, including signed hedging contributions. Percentages are unavailable when total risk is non-positive; they are not forced to sum meaningfully around zero.

The 14.61% XLE target contributes about 31.76% of recent HS99 ES. Gold also has a substantial tail component despite its defensive label. A small nominal weight is not the same as a small risk budget. Allocations at the same exposure and sample are compared with constraint violations and proposed loss-limit status, not a backfit winner.

Drift is a policy issue

Across the full active path, 2771 of 4,987 pre-return sessions drift beyond at least one target cap. Maximum single-ETF exposure is 45.14%. Targets remain controlled; continuous compliance is not claimed. The committee must approve whether natural drift is acceptable or a new policy is required, then validate any changed policy separately.

Trigger	Proposed response
HS99 ES >2% or VaR99 >1.5%	Review feasible redistribution and the risk budget; do not raise limits simply to pass
Drawdown >20% / crisis shortfall	Review spending, liquidity needs and ability to tolerate the loss path
Quarterly clustering test p<0.05	Investigate model dependence; consider a separately validated conditional-volatility extension
Target-cap drift / missing sessions	Escalate governance / data review; no silent clipping, filling or stale forecast substitution

12 / Reproduce, challenge and defend

What remains conditional

Human mandate approval and independent validation remain PENDING. Assumptions about spending, additional fees, future inflation, loss tolerance and continuous constraint enforcement are proposed. Sample means, geometric growth, regression alpha and stress history are not forecasts. No liability cash-flow optimisation or guaranteed capital protection is claimed.

The historical rule uses estimated covariance and approximate closing-price execution. Costs are a fixed turnover model; liquidity, market impact, tax, borrowing and intraday slippage are not fully modelled. References differ in mandate feasibility. Recent / full-window risk sensitivity is disclosed instead of hiding favourable sample choices.

Reproducibility

The repository includes the verified frozen inputs, exact pinned Python3.11 dependencies, pipeline, tests, actual CSV / JSON / PNG / SVG results and Streamlit app. Run **python scripts/launch.py** after installation. It opens the local dashboard without market-data downloads. Rebuild with scripts/build_final_results.py; figures and PDFs consume that persisted reference. Source versions and config hashes are embedded in results.

AI disclosure and required human review

Google Antigravity, Claude Code and OpenAI Codex materially assisted research, implementation, testing, analysis and report drafting; actual known tools and purposes are recorded in AI_DISCLOSURE.md. Automated tests do not substitute for the challenge's independent team review. The team must evaluate the recommendation, verify representative calculations, understand the code and sign the validation checklist before certifying this as its own decision.

Sources and primary references

[1] Federal Reserve Board, DTB3 discount-basis Treasury quote, via FRED: fred.stlouisfed.org/series/DTB3. Versioned evidence: ALFRED / archived H.15, paths and source URLs in data/reference/rf_evidence and audit/rf_research.

[2] SciPy official GPD distribution / support documentation: docs.scipy.org / scipy.stats.genpareto. Installed package versions are pinned in requirements.txt.

[3] Basel Committee (1996), supervisory backtesting framework: bis.org/publ/bcbs22.pdf. Educational traffic lights are distinguished from model approval.

Market / context: Yahoo Finance via yfinance; FRED CPIAUCSL (series link). Official competition requirements: Quant edge 1.0 Final Challenge 1.pdf, sections8-10. Full methodology, quality reports and evidence are in docs/sections and audit/. No proprietary data source is used.

Decision: CONDITIONAL ADJUST. Preserve the rule as a candidate; resolve the funding, tail budget and governance questions before investing.